

CYLINDER DISTRIBUTION PLATFORM.

Production-Grade LPG Delivery Backend

TEAM CYLDIST

Department of Computer Engineering

Member 1 - Lead Developer

Member 2 - Backend Engineer

Member 3 - DevOps & DB

Member 4 - Frontend & QA

[Your College Name] | Academic Year 2025-26

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INTRODUCTION

Digital LPG Distribution Reimagined.

A production-grade backend that transforms how LPG cylinders are booked, tracked, and delivered.

PROJECT IMPACT

3+

User Roles

30+

API Endpoints

10+

Core Features

2

Databases

KEY OBJECTIVES

Goals & Mission

1

DIGITIZE

Replace manual booking
with online + WhatsApp

2

TRACK

Real-time GPS tracking
& live customer chat

3

SECURE

JWT + RBAC + bcrypt
production security

4

SCALE

Docker + Kubernetes
ready deployment

Problem vs Solution



TRADITIONAL SYSTEM

Outdated - Manual - Inefficient

- ✗ Manual booking via phone calls
- ✗ No real-time delivery tracking
- ✗ Opaque inventory management
- ✗ Zero priority for emergencies
- ✗ Paper-based record keeping
- ✗ No customer communication
- ✗ Inefficient route planning

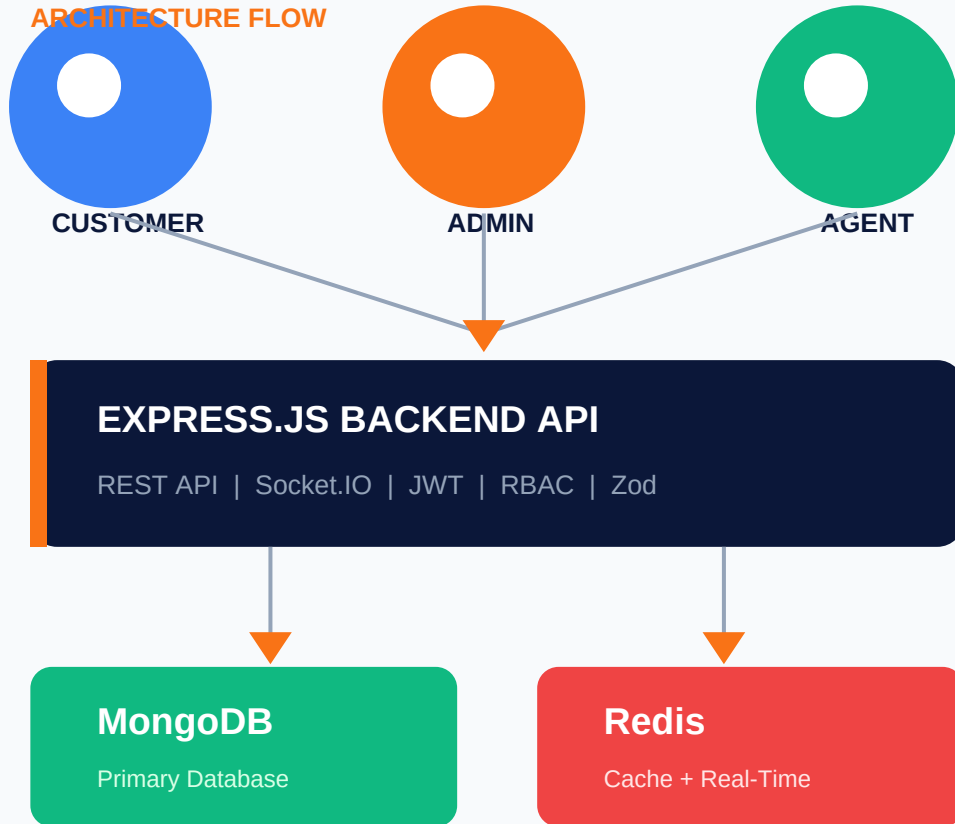


OUR DIGITAL SOLUTION

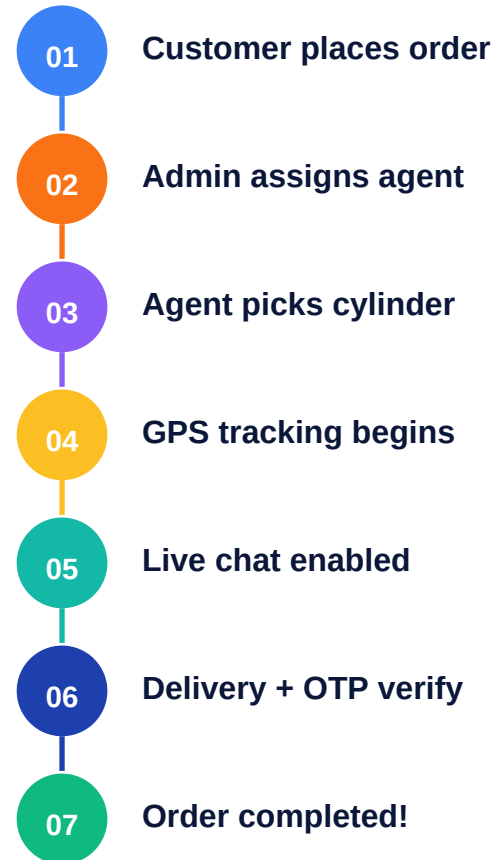
Modern - Automated - Scalable

- ✓ Online + WhatsApp booking
- ✓ Live GPS delivery tracking
- ✓ Automated inventory engine
- ✓ Crisis prioritization system
- ✓ Digital order management
- ✓ Real-time agent-customer chat
- ✓ Smart route optimization

Architecture & Workflow



ORDER LIFECYCLE FLOW



Modern Technologies

Built with industry-standard tools for scalability & reliability

N**Node.js**

Runtime
Engine

E**Express**

Web
Framework

M**MongoDB**

NoSQL
Database

R**Redis**

Cache &
RT Store

S**Socket.IO**

WebSocket
Layer

D**Docker**

Container
Platform

K**Kubernetes**

Orchestration
Engine

J**JWT**

Auth &
Security

SUPPORTING LIBRARIES & TOOLS

Security: bcrypt | Helmet | CORS | Zod | mongo-sanitize

Integration: Twilio | SendGrid | Razorpay | AWS S3 | Multer

Logging: Winston | Morgan | daily-rotate-file

Testing: Jest | Supertest | mongodb-memory-server

Unique Features

01

WhatsApp Booking

API-based booking via WhatsApp

02

Product Store

Buy accessories with cylinders

03

Live GPS Tracking

5-second GPS update intervals

04

Real-time Chat

Customer-Agent messaging

05

Crisis Engine

Emergency-aware allocation

06

Inventory Manager

Warehouse-level cylinder tracking

07

JWT + RBAC

Role-based access control

08

Multi-channel Notify

SMS / Email / Push alerts

09

Swagger API Docs

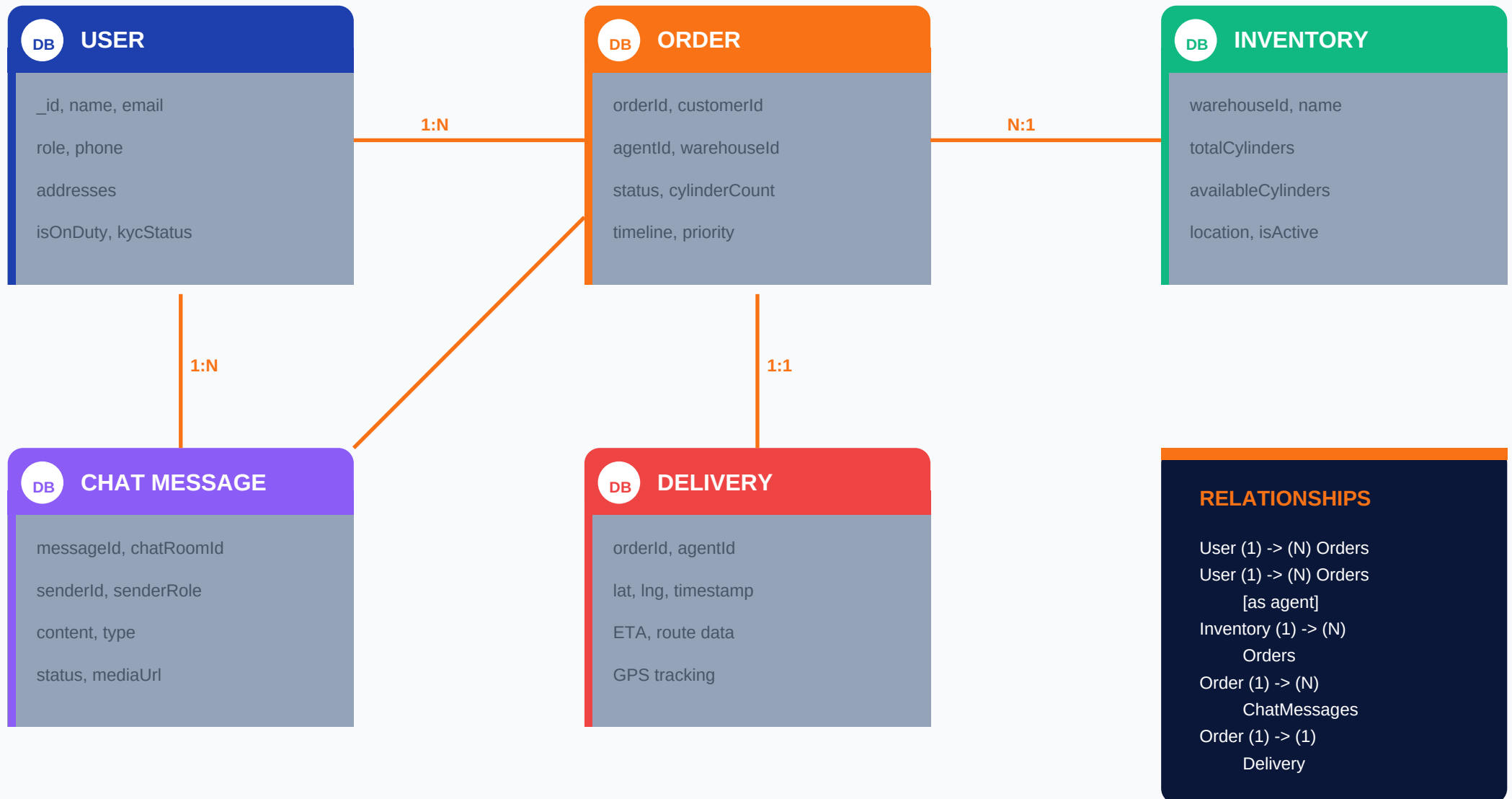
Interactive API testing UI

10

Docker + K8s

Production-grade deployment

Database Design & ER Diagram



Demo & Security

EXECUTION DEMO



1

Admin Dashboard

Order management & analytics

2

Swagger API DocsInteractive testing UI at `/api/v1/docs`

3

Live GPS Tracking

Real-time agent location updates

4

Agent-Customer Chat

WebSocket-based messaging

SECURITY ARCHITECTURE

S

MULTI-LAYER PROTECTION**JWT AUTH**

Access (15min) + Refresh (7d)
Token rotation + reuse detection

RATE LIMITING

100 req/15min global
10 req/15min for auth endpoints

ENCRYPTION

bcrypt 12-round password hashing
Helmet + CSP headers + HTTPS

RBAC

Role-based access control:
Admin | Customer | Delivery Agent

Helmet

CORS

HPP

Sanitize

XSS-Clean

Winston

Conclusion & Future Scope

PROJECT ACHIEVEMENTS

- ✓ Production-grade REST API
- ✓ Real-time GPS + Live Chat
- ✓ Crisis Prioritization Engine
- ✓ Complete Order Lifecycle
- ✓ Docker & Kubernetes Deploy
- ✓ Multi-layer Security (JWT/RBAC)
- ✓ WhatsApp Integration
- ✓ Swagger API Documentation
- ✓ Inventory Management
- ✓ Real-time Notifications

FUTURE ENHANCEMENTS

- AI**
Demand Prediction
ML-based consumption forecast
- APP**
Mobile App
React Native / Flutter
- PAY**
Online Payments
Razorpay / Stripe / UPI
- MAP**
Route Optimization
Google Maps integration
- IoT**
Smart Sensors
Cylinder gas-level IoT

THANK YOU!

Questions & Discussion Welcome